

Reducing Customer Churn at a Telecom Provider

A data-driven retention strategy projected to recover \$275K+ annually

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Executive Summary

- **The problem:** A telecom provider is losing \$1.67M annually to customer churn — 26.5% of customers leave each year, and churners pay 21% more on average than retained customers (\$74.44 vs \$61.27/month).
 - **The finding:** A single customer segment — month-to-month contract holders with fiber internet — drives **72% of all churn revenue** while representing just 30% of the customer base. 53% of all churn happens in the first 6 months of tenure.
 - **The recommendation:** A targeted retention program focused on month-to-month fiber customers in their first year — combined with first-6-month onboarding investment — could recover an estimated **\$275K+ in annual net business value** while requiring far less spend than broad retention campaigns.
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Business Context

For subscription-based businesses, customer churn directly erodes recurring revenue. Industry research consistently shows that acquiring a new customer costs **5–7x more than retaining an existing one**, making retention one of the highest-ROI investments a telecom can make.

This analysis examined 7,043 customers from a US telecom dataset (IBM Telco Customer Churn) to answer three questions:

1. **Who** is most likely to churn?
 2. **Why** are they churning?
 3. **What** can the business do, and what is the projected impact?
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Approach

The analysis combined exploratory data analysis, normalized SQL data modeling, and predictive machine learning across a unified pipeline. Customer demographic, subscription, service, billing, and churn data were loaded from CSV into a normalized 5-table PostgreSQL schema. SQL queries quantified the size and concentration of the churn problem across segments. A logistic regression model trained on engineered features achieved a **0.839 ROC-AUC** on a held-out test set, recovering 80% of churners at the optimal threshold. SHAP analysis confirmed the same drivers identified in exploratory analysis. A four-page Power BI dashboard built on top of the live database visualizes findings for operational use.

Key Findings

1. Contract type is the single largest predictor of churn

Customers on month-to-month contracts churn at **42.7%**, compared to **11.3%** for one-year contracts and **2.8%** for two-year contracts — a **15x gap** between the worst and best contract types. Month-to-month customers alone generate **86.9% of all churn revenue**, totaling **\$1.45M annually** out of the \$1.67M total. The two-year contract segment, while equally large in customer count, contributes just **3% of churn revenue** — essentially a non-problem.

2. One customer segment drives 72% of all churn revenue

Cross-segmenting by contract type AND internet service reveals a sharper picture: **month-to-month customers with fiber internet** (2,128 customers, just 30% of the base) churn at **54.6%** — more than half — and generate **\$1.21M in annual churn revenue**. This single segment alone accounts for **72.2% of total churn revenue**. By comparison, the bottom four segments combined account for less than 4%. This is a textbook Pareto pattern: a concentrated source of risk that responds to focused intervention.

3. Customer retention is fundamentally a first-year problem

Of customers in their first 6 months, **53% churn**. Retention jumps **17 percentage points** between months 6 and 12 — the single largest improvement in the entire customer lifecycle. By 4+ years of tenure, retention stabilizes above 80% and grows only marginally thereafter. **Investment in onboarding, early engagement, and proactive support during the first six months would address the company's largest retention leak**. Beyond 12 months, retention spend faces diminishing returns.

4. Churned customers pay more, not less

Contrary to the assumption that churn is dominated by price-sensitive low-value customers, **churners pay 21% more on average** than retained customers (\$74.44 vs \$61.27/month). This means the company is losing its highest-value customers, not just any customers. The lifetime value of saving these customers compounds: average monthly charges grow from \$54 to \$76 between months 6 and year 5+, a **39% revenue increase per customer** from tenure alone.

5. Predictive modeling validates and operationalizes the findings

Three models were trained and compared on the same data:

Model	ROC-AUC	PR-AUC	Recall on Churners
Logistic Regression	0.839	0.641	80.0%
Random Forest	0.821	0.624	50.5%
XGBoost	0.831	0.647	75.9%

The simplest model — logistic regression — outperformed both ensemble methods. This is itself a finding: the relationship between features and churn is largely linear and additive, so a more complex model adds operational cost without performance gain. The logistic model has the further advantage of full interpretability — each customer's risk score can be explained in business terms.

Cost-based threshold tuning identified an optimal classification threshold of 0.50, producing **\$55K in net business value on the held-out test set** (after retention program costs). Scaled to the full customer base, this represents approximately **\$275K in annual net value recovery** — roughly 17% of the \$1.67M churn loss, with significant upside if retention success rates exceed the conservative 30% assumption.

Recommendations

1. Contract conversion program for month-to-month fiber customers

Action: Launch a targeted retention offer for the 2,128 month-to-month fiber customers, with meaningful incentives (e.g., a 15% discount, locked-in pricing, or upgraded service tier) in exchange for a 1- or 2-year contract commitment.

Rationale: This single intervention addresses the segment driving 72% of churn revenue. Converting these customers from 54.6% churn (month-to-month fiber) to 19.3% churn (one-year fiber) would reduce segment churn by ~35 percentage points.

Estimated breakeven: At \$87/month average revenue, even a 15% discount preserves \$89/month in net revenue per converted customer. The math is overwhelmingly positive at any reasonable conversion rate above 5%.

2. First-six-month onboarding investment

Action: Build a structured first-six-month customer journey including a 30-day check-in, proactive support touchpoints at months 2 and 4, and a "completed onboarding" milestone reward.

Rationale: 53% of all churn happens in the first 6 months, and retention improves dramatically once customers cross the 12-month mark. The greatest leverage in the entire customer lifecycle is during the first year.

3. Service add-on attachment during onboarding

Action: Bundle online security and tech support as default add-ons (with opt-out) during the first 90 days, with discounted pricing.

Rationale: Customers without protective services churn at ~42%, while those with them churn at ~15% — a finding consistent across four different add-on types. Whether the services *cause* lower churn or merely *correlate* with more-committed customers, defaulting customers into protection creates either way.

4. Auto-pay conversion campaign

Action: Identify the customers paying via electronic check (the highest-churn payment method at 45.3%) and offer a one-time bill credit (\$10–\$25) for converting to auto-pay (bank transfer or credit card).

Rationale: Auto-pay customers churn at ~16% vs 45.3% for electronic check — a 3x gap. While payment method is likely a proxy for customer commitment, the conversion itself is a low-cost behavioral nudge that signals (and reinforces) commitment.

5. Deploy the predictive model for daily operational scoring

Action: Run the logistic regression model nightly on the customer base, surfacing the Critical and High tier list (currently 2,640 customers) to the retention team as a prioritized call list.

Rationale: Rather than relying on intuition or post-hoc analysis, the retention team can act on data-driven prioritization daily. The model's 80% recall on churners means a focused effort on the top-ranked customers

addresses the vast majority of at-risk revenue.

Projected Impact

If the recommendations above are implemented at conservative rates, the projected annual impact is approximately:

Recommendation	Annual Impact (Conservative)
Contract conversion (10% conversion of M2M fiber)	\$120K
First-6-month retention improvement (5pp churn reduction)	\$80K
Service add-on attachment (10% improvement in retention)	\$45K
Auto-pay conversion (15% behavioral nudge)	\$30K
Total estimated annual recovery	≈ \$275K

This represents a **~17% recovery of the \$1.67M churn problem**, achieved with focused intervention on roughly 30% of the customer base. Aggressive implementation, higher retention success rates, or compounding effects from multiple programs working in concert would push this number significantly higher.

The targeting precision matters: a broad retention spray-and-pray approach would burn most of its budget on customers who weren't going to leave anyway. The surgical approach concentrates spend on the segments where impact is highest.

Methodology Notes & Limitations

- Data source:** IBM Telco Customer Churn dataset (7,043 customers, anonymized). Findings should be re-validated against the actual telecom's data before operational rollout.
- Model performance:** ROC-AUC of 0.839 is strong for tabular data, but real-world performance depends on data drift, which should be monitored over time.
- Cost assumptions:** Retention offer cost (\$50/customer) and retention success rate (30%) are industry-typical but should be calibrated against the actual telecom's historical campaign data.
- External factors:** This analysis does not incorporate competitive dynamics (e.g., fiber pricing from competitors), service quality data (e.g., outage history), or macroeconomic factors. A production model would integrate these signals.
- Causation vs correlation:** Service add-ons correlate strongly with retention but causation cannot be confirmed from this dataset alone. A randomized rollout test is recommended before scaling the add-on attachment recommendation.

Supporting Materials

- Interactive dashboard:** [Power BI live link in README]
- GitHub repository:** github.com/Jaswanthi-A/churn-analysis
- Notebooks:** [notebooks/01_edu.ipynb](#) (exploratory analysis), [notebooks/02_modeling.ipynb](#) (predictive modeling)